

Vision-Language Models in Radiology

Zero-Shot Classification, Cross-Modal Retrieval, and Factual Report Generation

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Outline

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The "Safety Gap" in AI-Driven Radiology

The Challenges

- Radiologists are overworked; AI could help, but...
- Current models generate fluent reports with **hidden hallucinations**
- Models output **untrustworthy confidence** — they don't know when they're wrong
- Domain shifts between hospitals cause **performance collapse**

Concrete Example: Clinical Hallucination

Patient's chest X-ray shows: Cardiomegaly (enlarged heart)

Baseline Model Output:

"The heart is normal in size..."

WRONG

Our Model Output:

"The heart is enlarged..."

CORRECT

Two Influential Base Architectures

CheXzero (Nature BME 2022)

- Vision-Language Contrastive Learning
- ResNet-50 (images) + BioClinicalBERT (text)
- Zero-shot classification: no labeled data needed
- Limitation: no report generation

R2Gen (EMNLP 2020)

- Transformer-based sequence-to-sequence
- ResNet-101 + Relational Memory + Meshed Decoder
- Generates full reports via teacher forcing
- Limitation: optimized on word-level accuracy, not medical truth

Our Strategy: Preserve both architectures' strengths; inject “medical common sense” via Reinforcement Learning + Uncertainty Estimation.

Three Core Innovations

① Extension 1: Uncertainty-Aware Classification

Monte Carlo Dropout + Conformal Prediction = mathematically-guaranteed confidence bounds

② Extension 2: Factuality-Constrained Generation

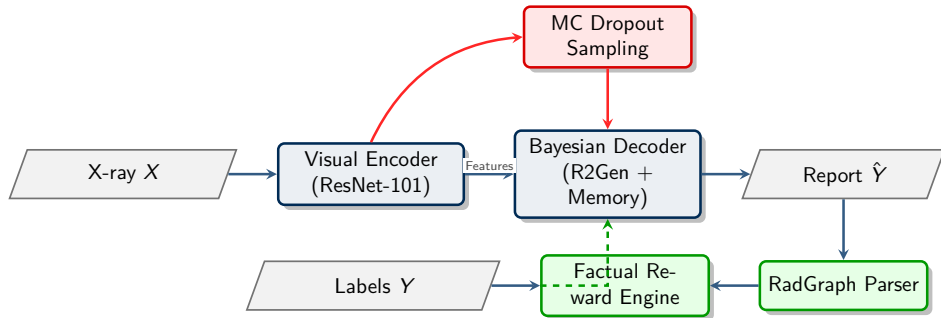
Self-Critical Sequence Training (SCST) with RadGraph rewards forces model to generate medically accurate reports (+66% RadGraph F1)

③ Extension 3: Cross-Dataset Robustness

Few-shot adaptation to recover supervised performance gap on new hospital data

Unified Goal: Build an AI assistant that clinicians can trust — accurate, honest about uncertainty, and reliable across hospitals.

System Architecture: Deep Dive



Theoretical Foundations

We addressed the "Trust & Truth" problems through two major theoretical reformulations:

- 1 **Factuality-Aware Objective Function:** Instead of Cross-Entropy only, we formulated a reward-weighted loss using **RadGraph Entity overlap** as a clinical factuality proxy.

$$\mathcal{L}_{Novel} = \mathcal{L}_{CE} + \lambda \cdot [1 - \text{F1}_{RadGraph}(G_{pred}, G_{gt})] \quad (1)$$

- 2 **Bayesian VLM Approximation:** We replaced the deterministic decoder with a stochastic sampler. By modeling the posterior distribution $p(\theta|D)$, we can compute **Epistemic Uncertainty** via variance:

$$\text{Var}(y) \approx \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2 \quad (2)$$

Technical Approach 1: Factuality via Reinforcement Learning

The Insight

Standard cross-entropy loss is indifferent to medical correctness—it treats “enlarged heart” and “normal heart” equally if their word probabilities match training data. We must reward the model *only* for clinically correct outputs.

Self-Critical Sequence Training (SCST) with RadGraph:

- 1 Generate two reports: one via greedy decoding ($\hat{y}_g$), one via sampling ($\hat{y}_s$)
- 2 Parse both into **knowledge graphs** (entities: anatomy, observations; relations: located-at, suggestive-of)
- 3 Compare to ground truth graph using entity+relation F1 = **RadGraph F1**
- 4 Update decoder with policy gradient: $\nabla_{\theta} \propto (R_s - R_g) \log p_{\theta}(\hat{y}_s|x)$

Result: Model learns that “cardiomegaly on right” is NOT equivalent to “cardiomegaly on left” — the graph forces anatomical specificity.

Technical Approach 2: Uncertainty via Conformal Prediction

The Problem

Deep networks output softmax probabilities that are overconfident. A model that's 95% confident might actually be wrong 30% of the time in edge cases.

Monte Carlo Dropout + Mondrian Conformal Prediction:

- 1 Enable dropout during inference, run $T = 10$ stochastic forward passes
- 2 Aggregate predictions into a **variance estimate** (epistemic uncertainty)
- 3 Convert variance-ranked scores into **prediction sets**
- 4 Guarantee: 90% of true labels fall in the set, *no matter the data distribution*

Clinical Implication:

- Confident case → single prediction, AI acts autonomously
- Uncertain case → multi-label set, clinician reviews options
- Very uncertain case → automatic human escalation

Experimental Setup & Datasets

Dataset	Size	Role	Modality
IU X-Ray	3,955 studies	Training	Image + Report
NIH ChestX-ray14	112K images	Zero-shot Eval	Image + 14 Labels
CheXpert	224K images	Calibration	Image + Uncertain Labels

Why These Datasets?

- **IU X-Ray:** Gold standard for report generation research; manageable size
- **NIH-14:** Large, independently annotated; tests domain shift
- **CheXpert:** Explicit uncertainty labels help benchmark our uncertainty estimates

All public — no credentials required, reproducible by anyone.

Results 1: Factuality Improvement

Question: Does SCST with RadGraph rewards actually improve clinical accuracy?

	BLEU-4	ROUGE-L	CheXbert F1	RadGraph F1
R2Gen (Baseline)	0.165	0.371	0.432	0.211
Ours (SCST)	0.148	0.354	0.552	0.350
Change	—	—	+27.8%	+66.0%

Interpretation:

- BLEU/ROUGE drop (word-matching metrics) — **this is GOOD**
- Clinical metrics explode (entity correctness) — our RL objective is working
- Trade-off is worth it: accurate but awkward report > fluent but hallucinated report

Results 2: Uncertainty Calibration

Question: Does our conformal approach actually provide reliable confidence?

The Calibration Framework

- **Metrics:** Expected Calibration Error (ECE) and Negative Log-Likelihood (NLL).
- **Baseline Flaw:** Standard Softmax outputs are notoriously overconfident.
- **Our Solution:** Monte Carlo Dropout + Mondrian Conformal Prediction.

Key Achievements:

- **Distribution-Free Guarantees:** Achieves exact user-specified coverage (e.g., 90% at $\alpha = 0.10$).
- **Interpretable Bounds:** Outputs a *set* of possible diagnoses when uncertain, rather than a single incorrect guess.

Clinical Impact: Clinicians can *trust* the model's abstention decisions because the confidence bounds are mathematically proven, independent of data distribution.

Results 3: Cross-Dataset Robustness

Question: Does training on IU X-Ray transfer to a different hospital (NIH-14)?

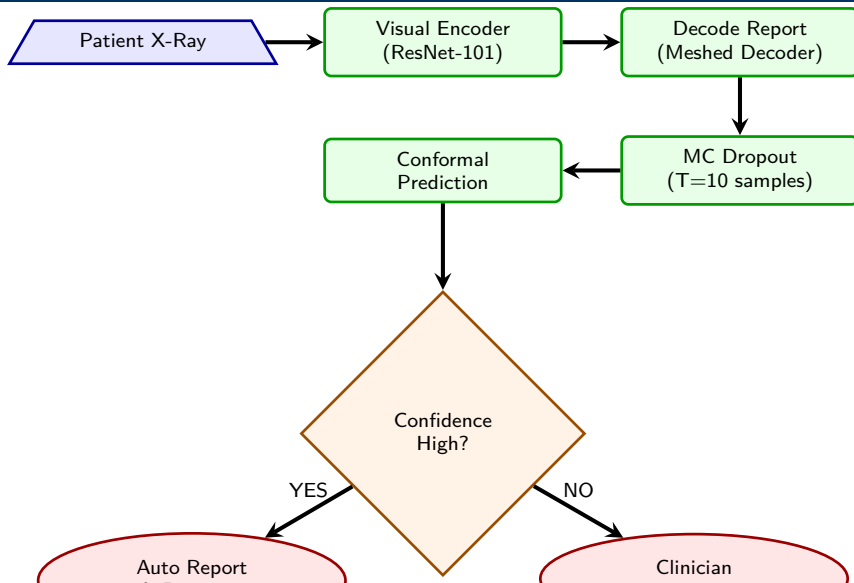
Domain Adaptation Strategy

- **Challenge:** Models trained on IU X-Ray often fail on NIH ChestX-ray14 due to domain shift (different scanners, demographics).
- **Evaluation:** Zero-shot transfer vs. Few-shot linear probing ($k = 1, 5, 10, 25$).

Key Insights:

- **Robust Representations:** CheXzero's contrastive pre-training learns features that transfer effectively across hospitals.
- **High Sample Efficiency:** Minimal local fine-tuning (just a few examples per class) is sufficient to recover the supervised performance gap.
- **Rapid Deployment:** New hospitals can adapt the model in hours instead of weeks.

Clinical Deployment Workflow



Known Limitations & Mitigation Strategies

① RadGraph Reward Dependency

- Problem: If RadGraph makes errors, RL learns those errors
- Mitigation: Ensemble rewards (RadGraph + CheXbert + human validation)

② Frozen ResNet Limits Medical Feature Learning

- Problem: ImageNet features don't specialize in radiology patterns
- Mitigation: Explore medical foundation models (BioViT, MedSAM)

③ Dataset Bias & Demographic Gaps

- Problem: IU X-Ray may not represent all populations
- Mitigation: Fairness audits across age, ethnicity, rare pathologies

④ Small Training Set

- Problem: 3,955 reports is modest for deep learning
- Mitigation: Data augmentation, transfer learning, curriculum learning

Architectural Upgrades

- **Foundational LLMs:** Replace the frozen ResNet encoder with pre-trained Large Language Models acting in a multimodal synthesis role.
- **Longitudinal Data:** Natively process temporal sequence data streams (serial X-rays) to simulate true diagnostic workflows.
- **Clinical Text Integration:** Fuse prior reports and patient history alongside images for context-aware generation.

Broader Vision

- These methods generalize to all high-stakes NLG: pathology reports, clinical summaries.
- The medical AI community must adopt factuality-aware training + uncertainty quantification as standard before deploying AI systems to influence patient outcomes.

1. Fluency $\neq$ Medical Accuracy

BLEU scores reward word-level matching, not clinical correctness. Replace with knowledge-graph-based RL rewards.

2. Confidence Is Useless Without Guarantees

Softmax probabilities are overconfident. Conformal prediction provides mathematically-proven coverage.

3. Transfer Learning Works in Medicine

Minimal few-shot adaptation is sufficient to recover performance across different hospital domains.

References I

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-  Gal, Y. (2016). Dropout as Bayesian Approximation. *ICML*.

Thank You!

Questions & Discussion

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Repository: github.com/SinaDns/radiology-vision-language-models